ENTRIX door controller — schematic

Rev A (draft) · 2026-11-01 · status: design proposal, not yet built or tested

Sheets

1. Power input and regulators
2. Microcontroller ESP32-C6
3. USB-C and programming header
4. 433 MHz receiver RXB61
5. Real-time clock and memory
6. Relay outputs
7. Door, exit-button and tamper inputs
8. LEDs and buzzer
9. Expansion header and RS485 (optional)

Main parts

- **U3** ESP32-C6-WROOM-1-N8: BLE 5 + Zigbee (802.15.4) + Wi-Fi 6, 8 MB flash
- **U5** RXB61 433.92 MHz superheterodyne receiver (RXB6 / RXB12 also fit electrically)
- **U1** XL7015E1 80 V buck → 5 V; **U2** AP2112K-3.3 LDO → 3.3 V
- **U6** PCF8563T RTC (SO-8) + BR2032 backup cell
- **K1** 5 V relay, 10 A, NO/COM/NC dry contact
- **U8, U9** PC817 inputs for door contact and exit button; tamper switch on J8

Connectors

- **J1** power 12–24 V AC/DC · **J5** relay 1 (COM/NC/NO) · **J7** inputs (DOOR/EXIT/COM)
- **J8** tamper · **J3** USB-C (service) · **E1/J4** 433 MHz antenna
- Optional (DNP): **J2** UART header, **J6** relay 2, **J9** expansion, **J10** RS485

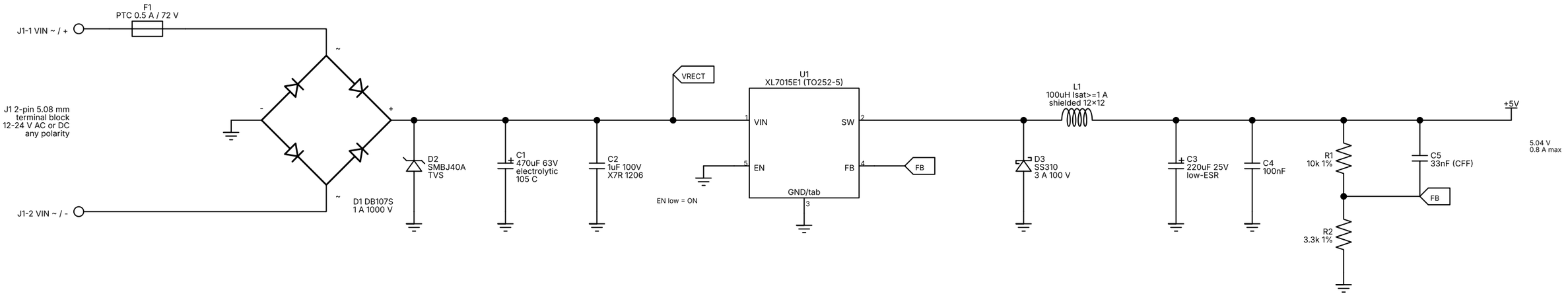
Before PCB layout, verify

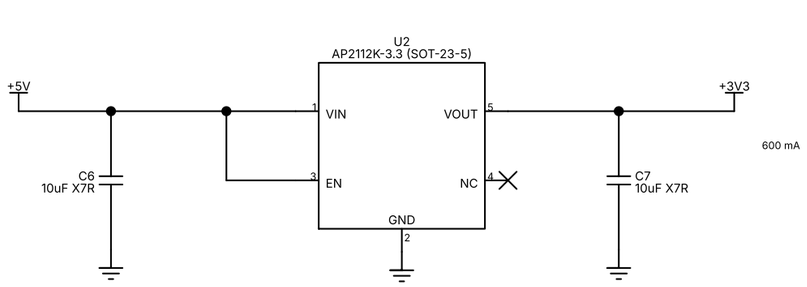
- RXB61 pin order and SHUT polarity on the actual module (no official datasheet found).
- ESP32-C6-WROOM-1 pin numbers against the current Espressif datasheet.
- XL7015 pin-out, C5 (CFF) and inductor value against the XLSEMI datasheet revision you buy.

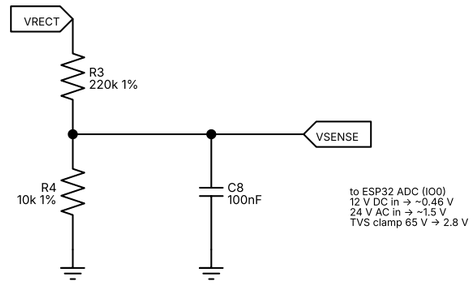
ESP32-C6 pin map

GPIO	Module pin	Net	Use
IO0	8	VSENSE	Input-voltage sense (ADC), power-fail detection
IO1	9	RF_DATA	433 MHz receiver data (via 10k/20k divider)
IO2	27	RELAY1	Relay 1 driver (Q1)
IO3	26	RELAY2	Relay 2 driver (Q2, optional)
IO4	4	EXP_IO4	RS485 DE/RE or LoRa M0 (10k pull-down)
IO5	5	EXP_IO5	LoRa M1 (expansion)
IO6	6	I2C_SDA	RTC (+ optional FRAM)
IO7	7	I2C_SCL	RTC (+ optional FRAM)
IO8	10	IO8_STRAP	Strapping pin, 10k pull-up
IO9	15	BOOT	SETUP / BOOT button
IO10	11	IN_DOOR	Door contact (opto)
IO11	12	IN_EXIT	Exit button (opto)
IO12 / IO13	13 / 14	USB_DN / USB_DP	USB-C (flashing, debugging)
IO15	23	-	Strapping pin, left open
IO16 / IO17	25 / 24	UART0_TX / RX	Debug console, header J2
IO18	16	IN_TAMPER	Lid tamper switch
IO19	17	LED_GRN	Green LED
IO20	18	LED_RED	Red LED
IO21	19	BUZZER	Buzzer driver (Q3)
IO22 / IO23	20 / 21	UART1_TX / RX	RS485 or LoRa module (optional)

Grey parts on the sheets are DNP (not fitted on the standard build). Netlist and BOM: netlist.md, bom.csv.



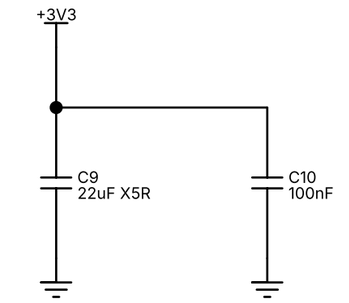




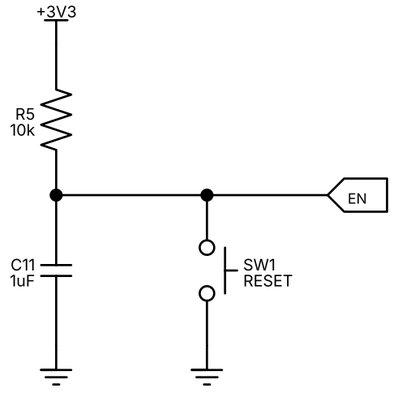
NOTES
1. Accepts 12-24 V DC (either polarity) or 12-24 V AC. 230 V sites: feed J1 from a sealed 12 V AC/DC module.
2. XL7015 must be the 80 V (or 100 V) version: 24 V AC rectifies to ~34-37 V; the SMBJ40A clamps surges at ~65 V.
3. Vout = 1.25 V x (1 + R1/R2) = 5.04 V. Keep the SW / D3 / L1 / C3 loop short; C2 right at U1 pin 1.
4. Use 105 C electrolytics (C1, C3): the box gets hot in sunlight.

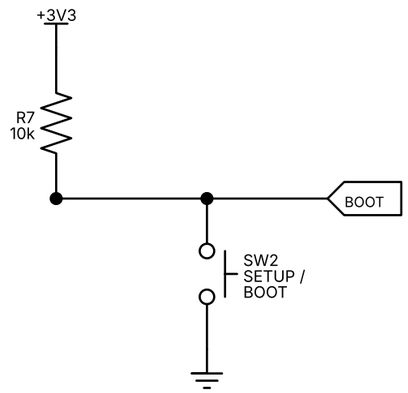
ENTRIX door controller
Power input and regulators

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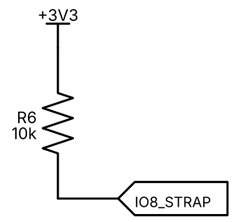


Place C9, C10 next to U3 pin 2 (3V3)

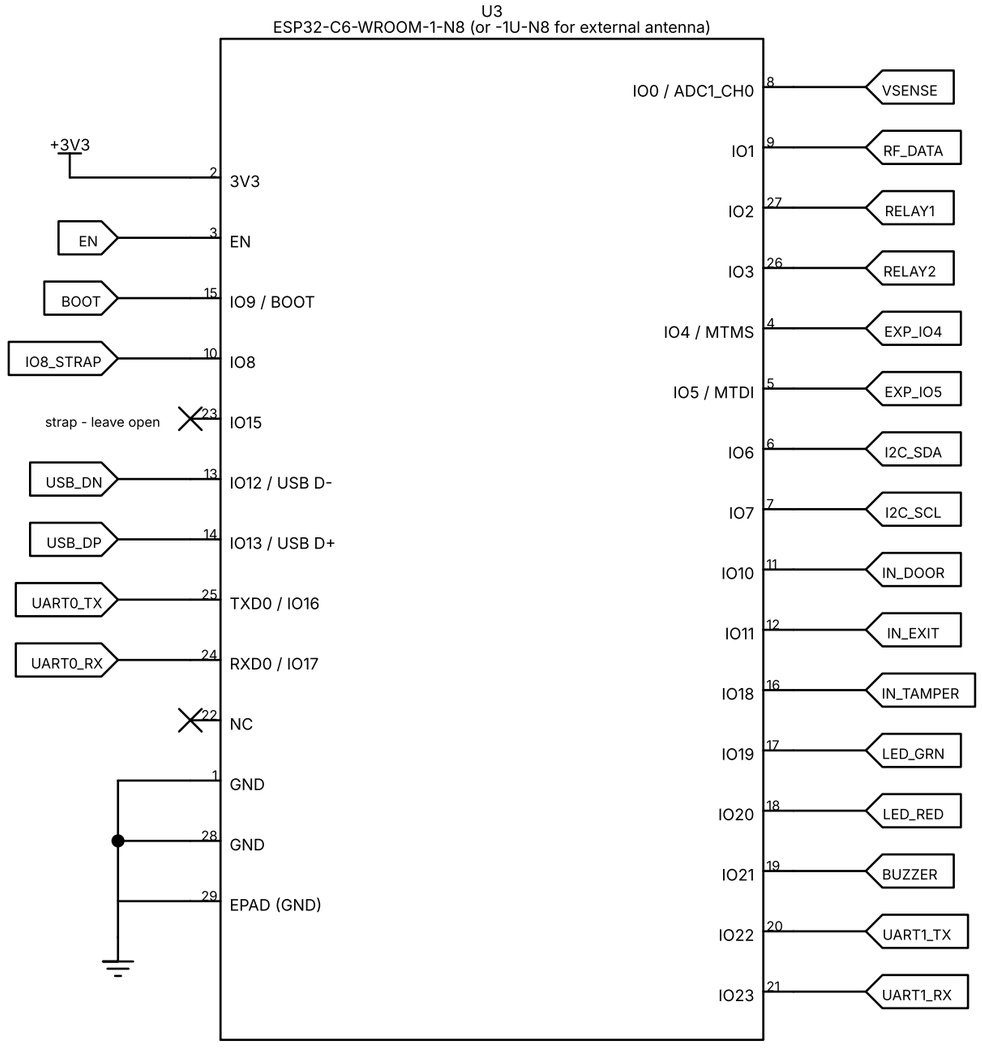




Hold SW2 at power-up = download mode.
Press in normal use = admin/setup mode.

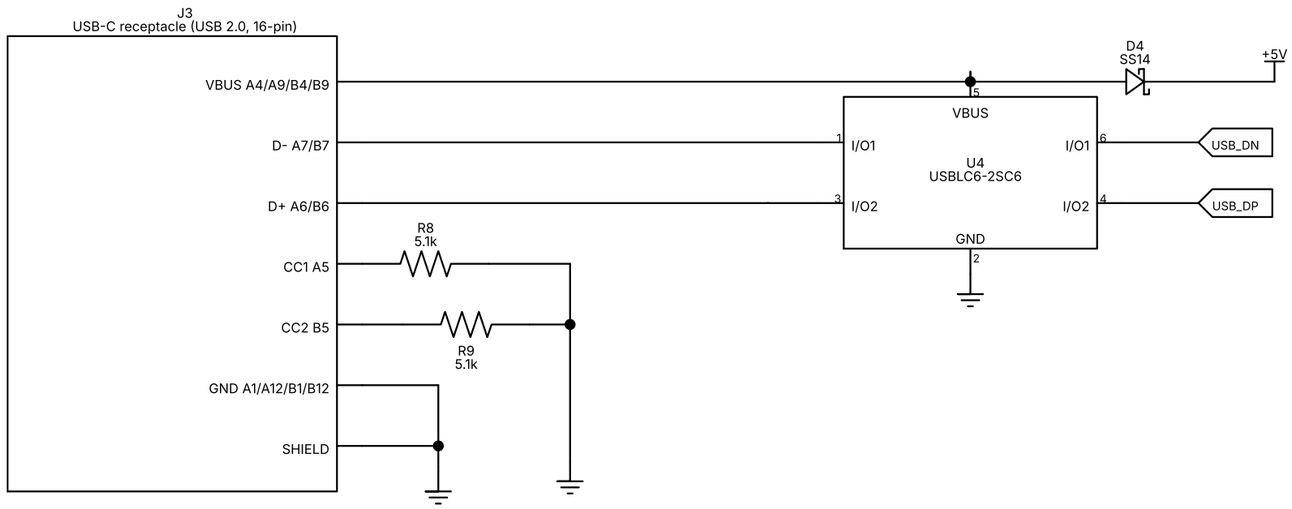


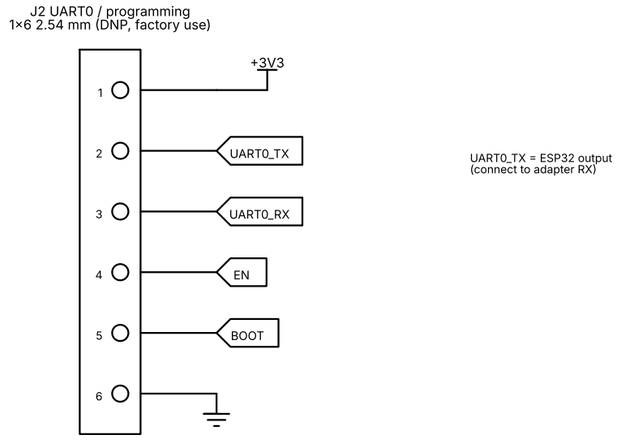
IO8 must be high for serial/USB download



NOTES
1. Pin numbers per ESP32-C6-WROOM-1 datasheet. Check against the current Espressif datasheet before PCB layout.
2. Strapping pins: IO8 (pull-up), IO9 (pull-up + button), IO15 (open), IO4/IO5 (expansion; IO4 has a 10k pull-down on sheet 9).
3. Keep the PCB-antenna end of U3 over the board edge with no copper underneath, away from the relay and XL7015.
4. Use the -1U variant + u.FL pigtail when the box is on a metal gate or Zigbee range must be extended.

ENTRIX door controller
Microcontroller ESP32-C6



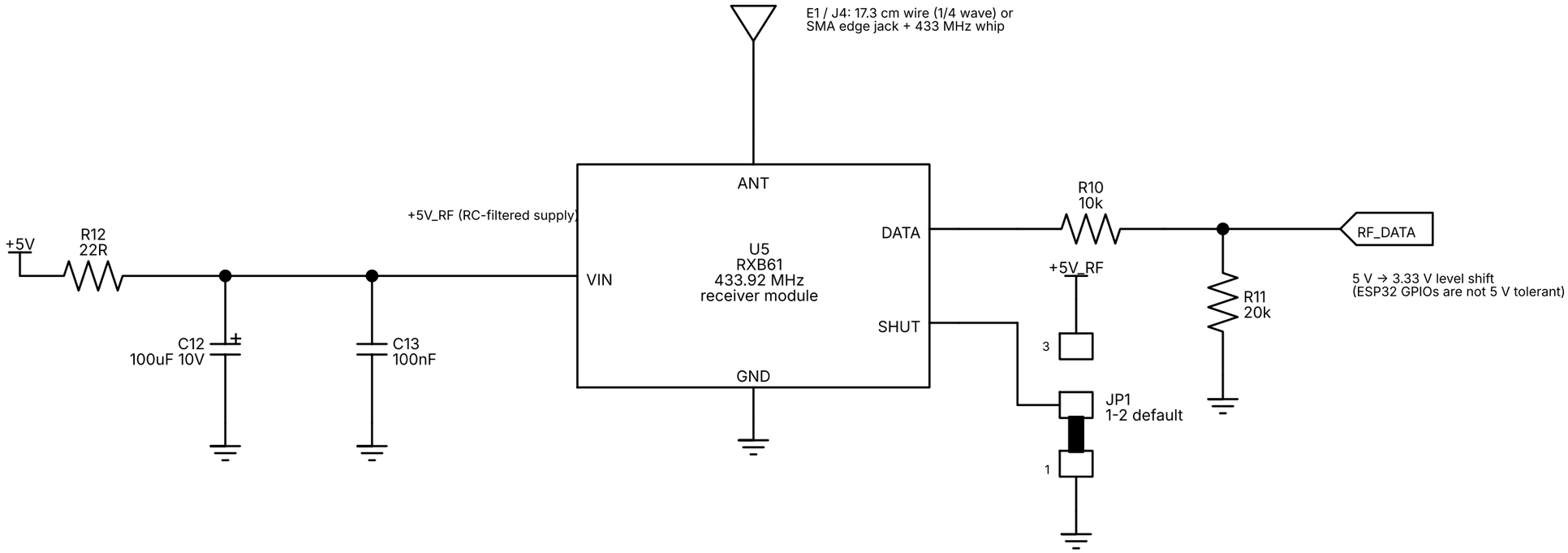


UART0_TX = ESP32 output
(connect to adapter RX)

NOTES
1. USB-C is for flashing and debugging through the ESP32-C6 built-in USB-Serial-JTAG. It is inside the box, not used in the field.
2. D4 lets the board run from USB on the bench (relay coil and RF receiver work at ~4.6 V). R8/R9 make the port a USB device (sink).
3. J2 is an alternative UART flashing port (needs a 3.3 V USB-UART adapter). Hold SETUP/BOOT while resetting to enter download mode.

ENTRIX door controller
USB-C and programming header

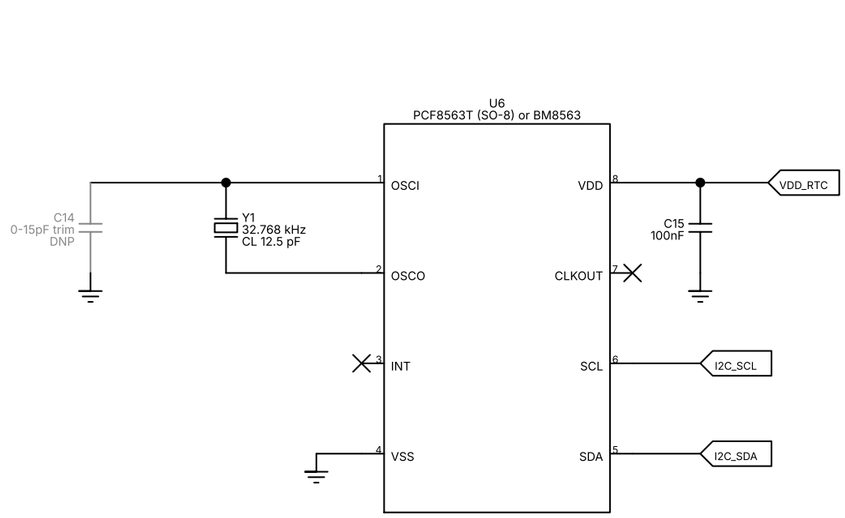
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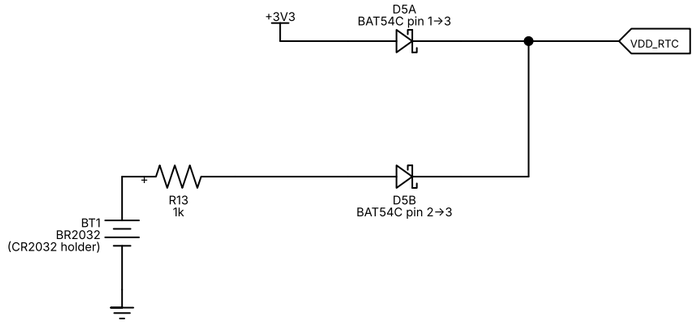


NOTES
1. RXB61 has 5 pins: VIN, GND, DATA, ANT, SHUT. No official datasheet was found: confirm the pin order on your module's silkscreen before PCB layout, and check the SHUT polarity (JP1 1-2 = SHUT to GND = receiver ON for active-high shutdown; use 2-3 if your module is active-low).
2. RXB6 / RXB12 / SRX882 can be used instead: same VIN, GND, DATA and ANT signals, different footprint (no SHUT pin → leave JP1 open).
3. Order the 433.92 MHz version (315 MHz in North America). Keep the receiver and its antenna away from the XL7015 (L1, D3) and relay K1.
4. Supply is 5 V for best sensitivity; R12/C12 filter the buck ripple. DATA is divided to 3.3 V by R10/R11.

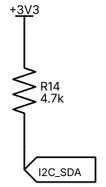
ENTRIX door controller
433 MHz receiver RXB61

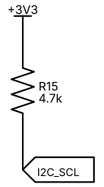
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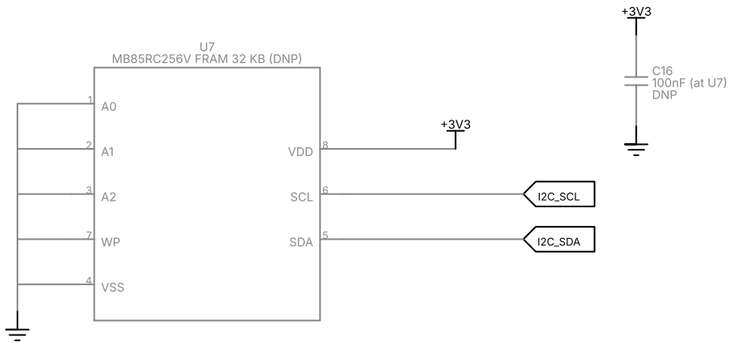




D5 = one BAT54C (common cathode, SOT-23).
R13 limits reverse current into the cell.



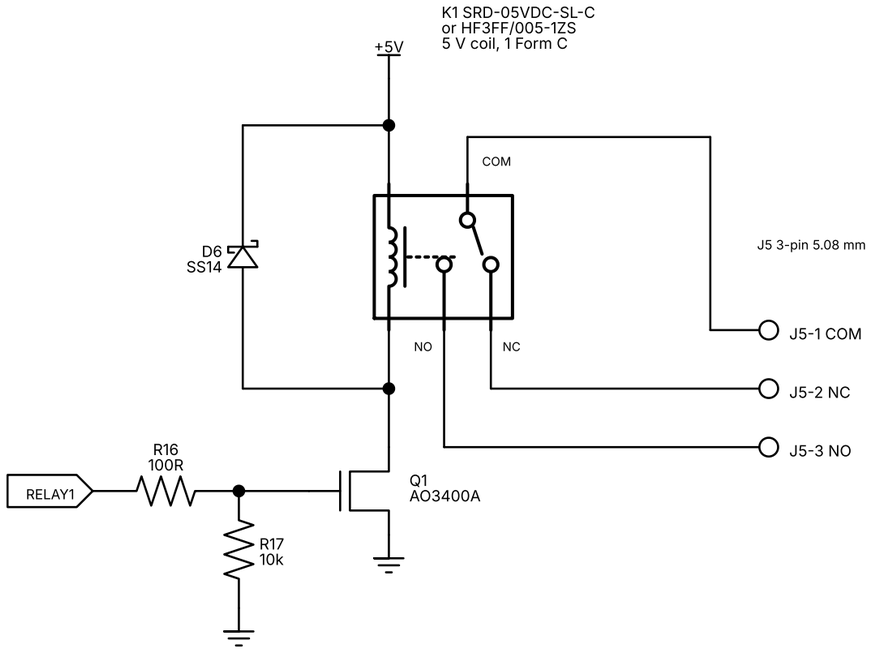


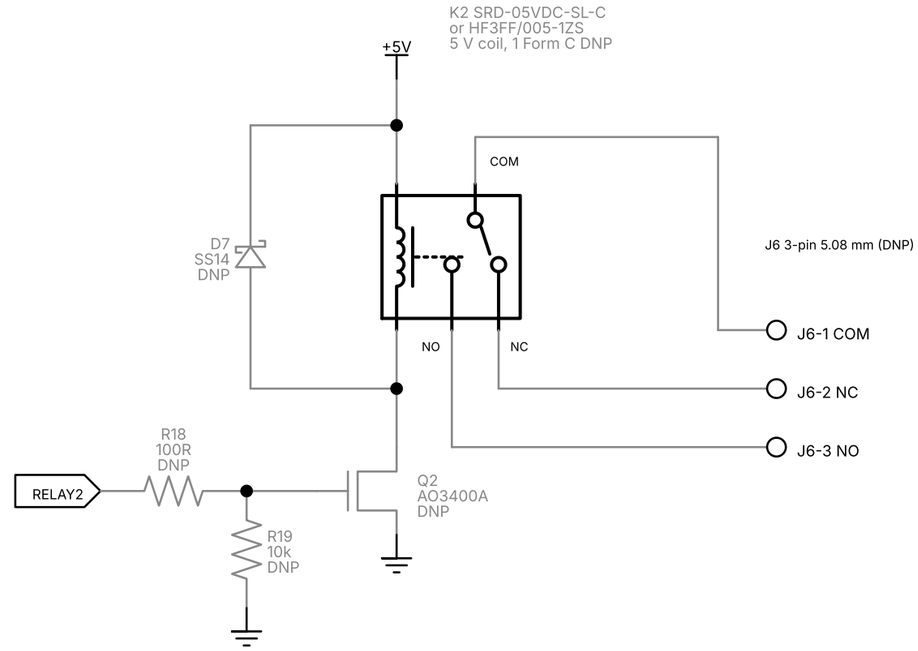


NOTES
1. PCF8563T is the PCF8563 in the SO-8 package (T = SO-8, TS = TSSOP-8, BS = HVSON-10); /5 or /F4 suffixes are revision/packing only. BM8563 (SO-8) is a pin-compatible lower-cost alternative. I2C address 0x51 (RTC), 0x50 (FRAM, optional).
2. The firmware checks the PCF8563 VL (voltage-low) flag at boot; if set, time is invalid until an admin or the gateway sets it.
3. BR2032 (Li/CF, -30…+85 C) instead of CR2032 for an outdoor box. RTC draws ~0.25 uA → the cell lasts many years.
4. Keep Y1 within 5 mm of U6 with a ground guard ring and no signals routed under it.

ENTRIX door controller
Real-time clock and memory

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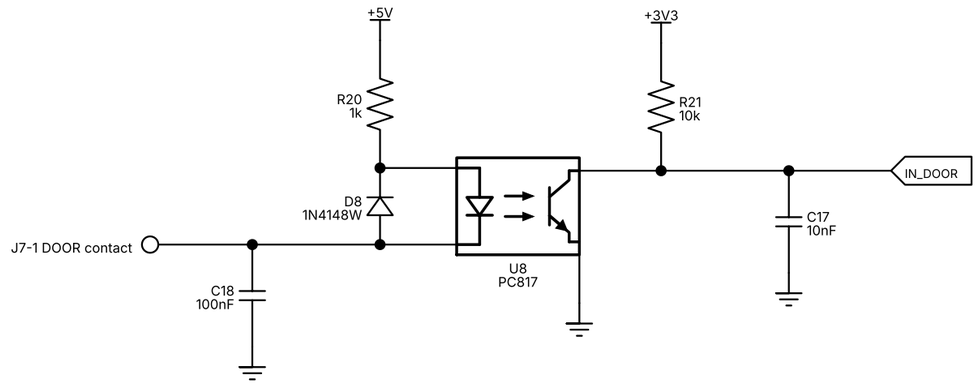


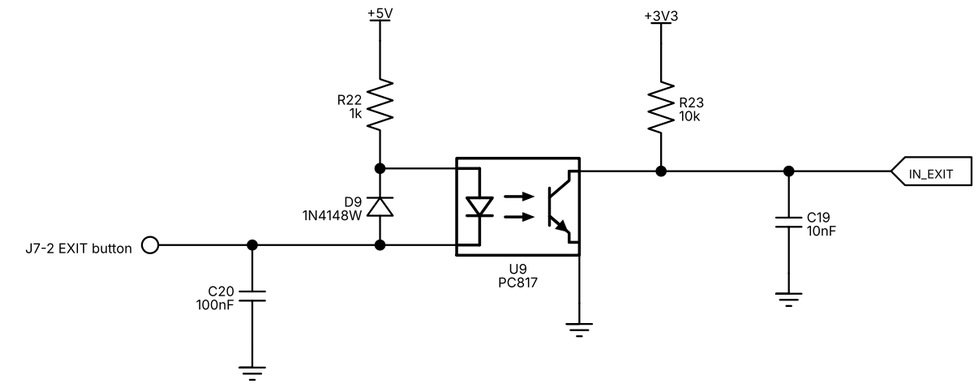


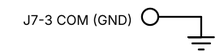
Contacts: 10 A 250 V AC / 30 V DC (relay rating). Potential-free (dry) contact.

NOTES
1. R17/R19 hold the MOSFET off while the ESP32 boots, so the door can never open during reset or power-up.
2. Strike / bolt (fail-secure): PSU+ → COM, NO → lock+. Maglock (fail-safe): PSU+ → COM, NC → maglock+.
 Gate / garage / sliding-door operator: COM + NO across its "push-button / open" input (0.5-1 s pulse).
3. Fit a flyback diode across DC lock coils (or a varistor across AC loads) at the lock.
4. If the relay ever switches 230 V, keep >= 6 mm creepage + a PCB slot between contacts and low-voltage circuits.
5. Channel 2 (K2, Q2, R18, R19, D7, J6) is optional: second door, roller shutter up/down, siren.

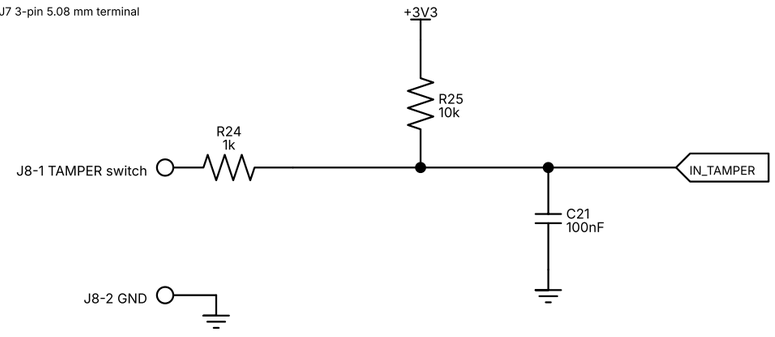
ENTRIX door controller
Relay outputs







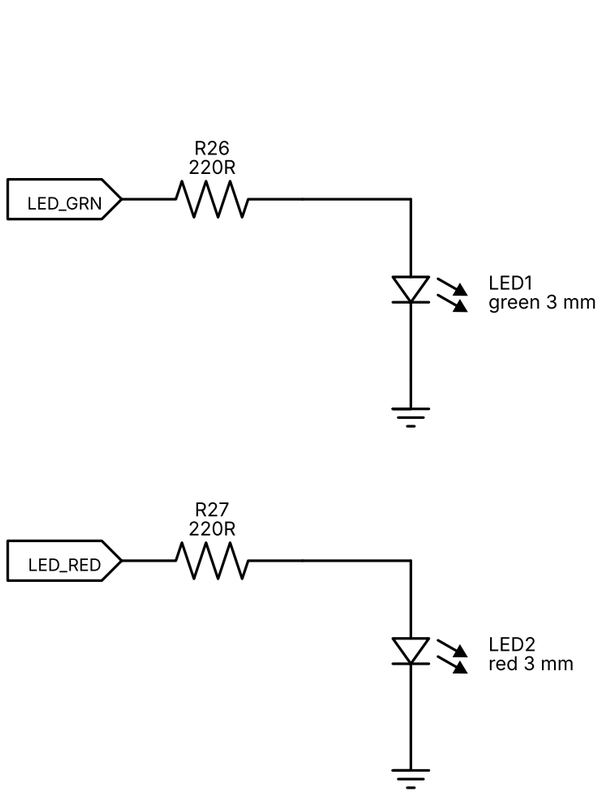
J7 3-pin 5.08 mm terminal

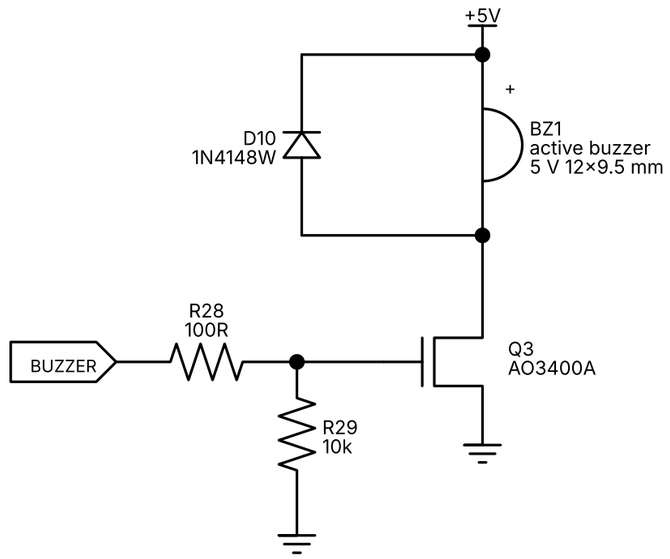


J8 2-pin JST-XH to lid micro-switch

NOTES
1. Contacts close to J7-3 (COM). Closed = opto LED on (~3.8 mA wetting current) = GPIO reads LOW.
2. Door contact: reed switch, closed when the door is shut. Exit button: normally-open push button.
3. Tamper: micro-switch pressed by the closed lid (closed = LOW). Lid opened or wire cut = HIGH = tamper event.
4. J7-3 is board ground, which is NOT isolated from J1. Do not connect J7-3 to the lock power supply.

ENTRIX door controller
Door, exit-button and tamper inputs





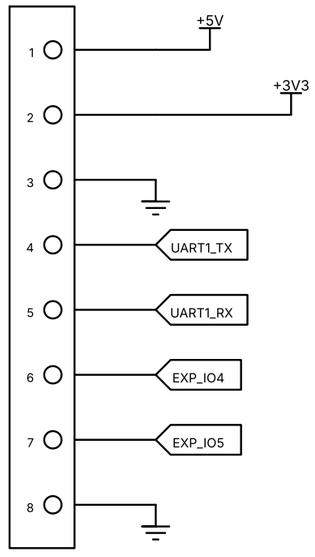
LEDs sit under a light pipe in the lid
(or wire them to an external indicator).

NOTES
1. Green = access granted, red = denied / tamper, alternating = admin mode. The buzzer gives a short beep on every remote press.
2. Use an ACTIVE buzzer (built-in oscillator): the firmware only switches it on and off.

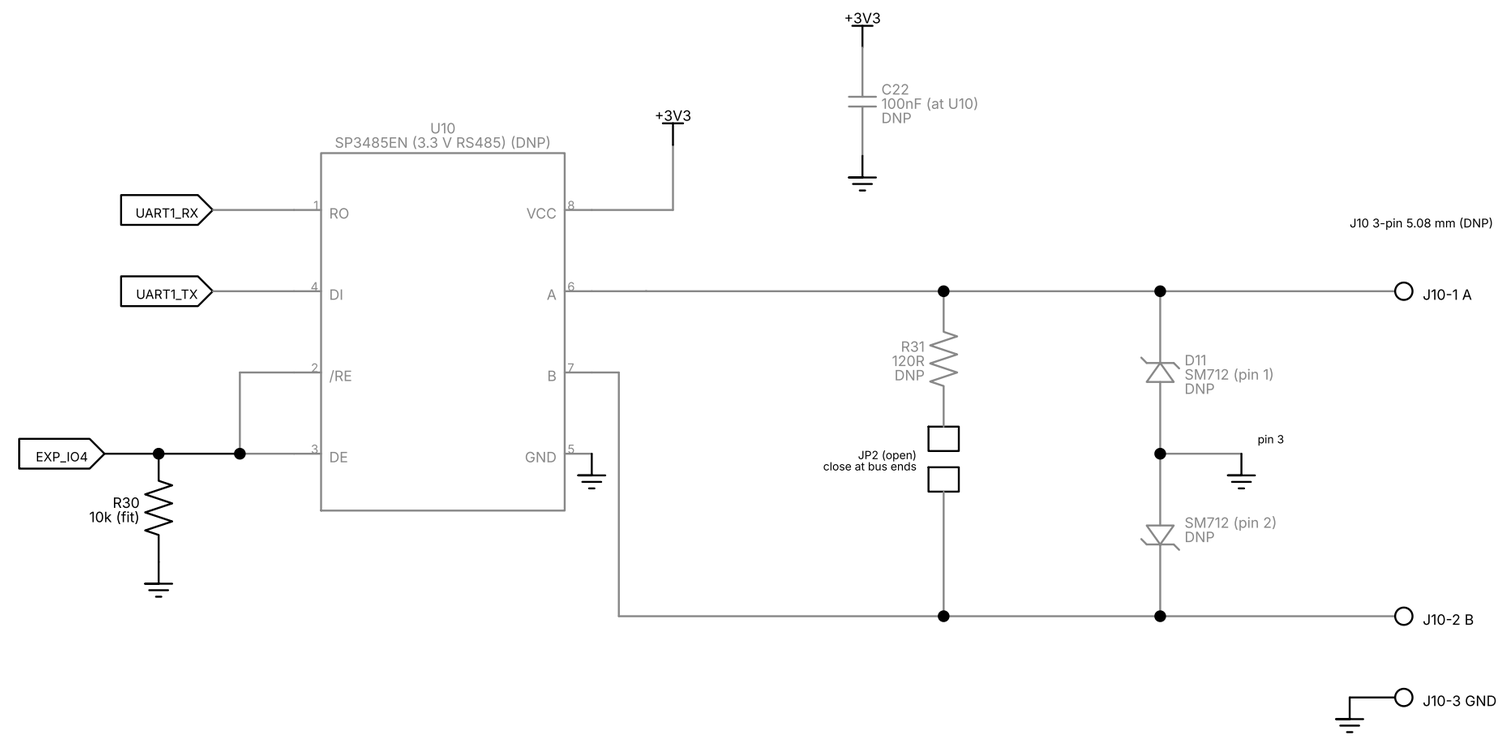
ENTRIX door controller
LEDs and buzzer

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J9 expansion 1×8 2.54 mm (DNP)
e.g. UART LoRa module E22-900T22S



LoRa: M0 = EXP_IO4, M1 = EXP_IO5.
Use 868/915 MHz modules, never 433 MHz.



NOTES
1. Everything grey is DNP (not fitted). Fit it only on units that need it. R30 is always fitted: it keeps IO4 low (RS485 = receive).
2. UART1_RX can have only one driver: fit EITHER U10 (RS485) OR a module on J9, never both.
3. Close JP2 (120 R termination) only on the two units at the physical ends of the RS485 cable.

ENTRIX door controller
Expansion header and RS485 (optional)

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